

Task: Binary Classification on UCI Adult (Census Income) Dataset

Objective: Predict whether an individual earns >50K/year for targeting a premium financial product.

1. Problem Framing

Business Problem: A financial institution wants to target high-income individuals (>50K/year) for a premium product. Marketing budget is limited, so we need to avoid wasting resources on low-income individuals.

Target Variable:

- Positive class (1): income >50K
- Negative class (0): income ≤ 50K

Base Rate: 23.93% of individuals earn >50K.

2. Chosen Metric

Primary Metric: Precision (for positive class)

- To minimize false positives (avoid contacting low-income individuals).
- Secondary metrics: Recall, F1, ROC AUC.

Justification: Precision is most aligned with the business goal — we want to be confident that when we contact someone, they are likely to be a high earner.

3. Baseline Results

Two simple baselines were implemented to establish minimum performance benchmarks.

Baseline	Accuracy	Precision	Recall	F1	ROC AUC
Majority Class (always predicts $\leq 50K$)	76.07%	0.00	0.00	0.00	0.50
Rule-Based (education-num $\geq 13 \rightarrow > 50K$)	75.30%	48.44%	49.70%	49.06%	0.6653

Observation: The rule-based baseline (education-num ≥ 13) is better than majority class, but still limited. Any complex model must outperform these baselines.

4. Error Analysis

False Positives (FP): 477 cases — model predicted $> 50K$, but actual was $\leq 50K$.

- Average age: 45.4 years
- Average education-num: 11.8
- Average hours-per-week: 47.7

False Negatives (FN): 949 cases — model predicted $\leq 50K$, but actual was $> 50K$.

- Average age: 42.6 years
- Average education-num: 10.2
- Average hours-per-week: 43.6

Key Patterns:

- False Positives tend to be older, work more hours, and have slightly higher education.
- False Negatives are younger, work fewer hours, and have lower education.

Feature Issues to Fix:

1. Missing values in workclass, occupation, native-country.
 2. Skewed features: capital-gain (75% zeros), capital-loss.
 3. High-cardinality categoricals: occupation, native-country.
 4. Outliers in hours-per-week (max 99 hours).
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End of Summary